Cloud computing fundamentals

1.

cloud computing
computing sewices delvived over Internet

means

Defination: Servers storage d atabases networking Software available online.

has

charecteristics:
on demand acess Board network access resources polling elasticity measure service

delivers

service delivery
Resource supplied when needed, usage can be measured.

Includes

on demand network access: users request resources themselves and access the from phones, laptop or pc's

Includes

polling+clasticity
shared provider resource capcity grows or shrinks with demand

Includes.

measured service
usage is monitered customs may pay for what they we.

Benefits: lower upfront coit access anywhere quict setup scability. lesshard ware maintanence

limitations; need internet privacy/security corceins passible down time provider depence ongoing tces

2.

mainframe computing.
large central computers served many users

Personal computer: computing become available to individuals

Server–client systems: clients requested data and services from servers.

Distributed systems: multiple connected computer shared work loads and resources

virtuliyation: one of physical server could run multiple virtual machine

modern cloud platforms: on-demand infrastructure platforms and software delivered online.

Hardware evolution:
more powerful servers
Storage and networking

software evolution
webapps, APIs and automation

Reasult:
flexiable
Scalable
services

3.



Responsibility shifts towards the user ← IaaS

paas/saas ⟶ more managed by provider.

4.

cloud development + scalability

cloud services are developed, connected and adjusted to

virtuligation

Abstracts physical hardware into virtual sevices

uses

Hypervisor

software that create and manages vms

supports

Abstraction

sperates workloards from s.specific physical hardsoare.

support

creates

virtual machine

Isolated - software baled computer with os and apps

allows

Resource polling

- shares cpu, memory and storage acess

Improves

web services

Allow applications to communitate over a network

API

Defined inter for software communication

serves

SOAP

protocol used xml messages.

or

Rest

Arch style commonly uling HTTP

consumer

Appor client requests a sevice

requet

communication

deveployment

provider allocate compute

Concumer access through Api of web apps

leads to

scalability + claiticity

Add / move virtual resource and adjust capacity with work load

leads to

Reacult: Efficient resource uce, connected Sevices and flexible cloud applications

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